

Belief-Augmented Reinforcement Learning for Pick-and-Place Under Observation Noise and Occlusion

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Abstract—Robotic pick-and-place under partial observability is challenging because sensor noise and occlusion corrupt or eliminate target state information, turning a standard MDP into a POMDP. Three solution strategies are investigated on the SO-ARM101 6-DOF arm in a MuJoCo environment featuring distance-dependent wrist camera noise and two occluding distractor blocks: (i) plain PPO with a stale-sensor observation model, (ii) belief-augmented PPO that feeds a bootstrap particle filter’s mean and uncertainty estimate directly into the policy observation, and (iii) POMCP-MCTS online planning as a stretch goal. The central finding is that particle filter belief augmentation provides a statistically meaningful advantage in the moderate-noise regime, peaking at +24% task success at $\sigma_{\text{obs}} = 10$ mm, but is outperformed by plain PPO at very low noise ($\sigma_{\text{obs}} = 3$ mm, −30%) where raw observations are already accurate and filter initialization overhead dominates. Both methods degrade at extreme noise (30 mm) due to filter impoverishment and stale-sensor failure respectively. The crossover near $\sigma_{\text{obs}} \approx 8$ mm is theoretically coherent: it marks the point at which filtering benefit first exceeds convergence cost, and is consistent with the signal-to-noise characteristics of a bootstrap particle filter with $N = 300$ particles.

I. INTRODUCTION

Robotic manipulation in unstructured environments requires acting reliably despite imperfect sensing. Real cameras return noisy detections, distractors occlude the target, and physical contact changes the scene in ways the robot cannot directly observe. Treating such settings as fully observable MDPs (as most deep RL work does) ignores the fundamental epistemic uncertainty in the system and can lead to brittle policies that fail when sensor quality degrades or objects move out of view.

This work examines a concrete instance of the problem: pick-and-place of a target LEGO block in the presence of two occluding distractor blocks, using only a wrist-mounted camera whose noise scales with distance to the target. Three distinct uncertainty sources are present simultaneously. *Sensor noise*: the wrist camera returns a Gaussian-corrupted position estimate whose standard deviation grows linearly with the end-effector-to-target distance, so the agent must approach the target to improve its own observations. *Occlusion*: when a distractor block passes between the camera and the target, the observation is entirely absent; the agent must remember and reason about the last known position. *Distractor identity*: because the auto-grasp threshold (10 mm) is comparable to the

inter-block spacing (~ 17 mm mean), the agent risks grasping the wrong block if it navigates to a stale or incorrect position estimate. Together these make the task a POMDP rather than an MDP: the true target pose is hidden, and observations are noisy, intermittent, and corrupted by irrelevant objects.

Two learned policy approaches are compared. *Plain PPO* treats the problem as memoryless: it receives the last visible wrist observation (replayed during occlusion) and learns to act on this noisy signal directly. *Belief-augmented PPO* maintains a bootstrap particle filter over the target’s (x, y, θ) pose and augments the policy observation with the filter’s mean μ and standard deviation σ , giving the policy an explicit uncertainty signal it can condition its behavior on. Both policies are trained with proximal policy optimization (PPO) [4] and evaluated across eight observation noise levels (3–30 mm). As a stretch goal, POMCP-MCTS is also evaluated [6] online planning with the simulator as a generative model.

The main finding is that the benefit of belief augmentation is strongly noise-regime-dependent. Belief PPO wins or ties at six of the eight noise levels tested and achieves its largest gain (+24%) at $\sigma_{\text{obs}} = 10$ mm, a regime where the particle filter converges quickly yet provides meaningful position smoothing. At very low noise (3 mm), however, plain PPO outperforms by 30%: the raw wrist observation is already accurate, and the filter’s initialization spread induces unnecessary caution in the policy. This noise-regime dependence is the central empirical contribution of the paper.

Departures from the proposal. Three aspects of the final system differ from the original proposal, each for concrete technical reasons. First, the proposal specified *episodic* observation noise (a different σ_{ep} drawn per episode during training) to produce a single generalist policy robust to all noise levels simultaneously. In practice, training with randomized noise destabilized learning: the particle filter’s initialization spread (set proportionally to σ_{ep}) varied wildly across episodes, preventing the policy from learning a consistent relationship between σ and appropriate caution. Training at a fixed σ_{ep} and evaluate the trained policy across all noise levels, which cleanly isolates the noise-regime structure. Second, the POMCP stretch goal was proposed with a *learned* world model (trained on rollouts from the belief PPO policy). Learning a faithful transition model for a MuJoCo contact-

rich environment within project scope is impractical; the simulator is used directly as the generative model, which is the standard approach in model-based POMDP solvers and does not change the algorithmic comparison. Third, the proposal described explicit *active perception*, rewarding the agent for moving to reduce uncertainty (decrease σ) as a separate reward term. This was implemented and evaluated; it failed because the information-gathering reward conflicted with the approach reward, causing the policy to wander away from the target to reduce occlusion rather than approaching to grasp. Removing the explicit reward and relying solely on the task reward produced a policy that still learns implicit uncertainty-conditioned behavior, as evidenced by the noise-sweep crossover.

II. BACKGROUND AND RELATED WORK

POMDPs for robotic manipulation. The partially observable Markov decision process (POMDP) framework [1] provides the formal foundation for decision-making under state uncertainty. In manipulation, partial observability arises from sensor noise, occlusion, and contact uncertainty. Koval et al. [2] show that contact interactions can be modeled as belief-space transitions, enabling planning over distributions of object poses rather than single estimates. This work addresses the simpler but practically important case of visual observation noise and occlusion, without the added complexity of contact-induced uncertainty.

Belief-state representations. Maintaining a belief over the hidden state is the key challenge in POMDP solving. The bootstrap particle filter [3] represents belief as a weighted set of samples and supports arbitrary non-Gaussian distributions, making it well-suited to manipulation tasks where occlusion can render observations completely absent. Kalman-family filters (KF, EKF, UKF) are computationally cheaper but assume unimodal Gaussian beliefs; this assumption breaks when the target has been occluded long enough that the true position could lie anywhere in a region, creating a diffuse, multi-modal belief that a single Gaussian cannot represent. In this setting, distractors can occlude the target for 50+ consecutive steps, making the particle filter the appropriate choice.

Deep RL for manipulation. Proximal policy optimization [4] is the de facto standard for continuous-action robotic RL due to its stability and ease of tuning. Reward shaping is known to be critical and treacherous: Ng et al. [5] show that only potential-based shaping preserves the optimal policy, while arbitrary dense rewards can introduce spurious local optima. Five distinct classes of reward-hacking exploit were discovered during development (per-step proximity farming, per-step gripper clamping, grasp-attempt spamming, fine-positioning hovering, and information-gathering conflicts), each illustrating a different failure mode of ad hoc shaping.

Online POMDP planning. Silver and Veness [6] introduced POMCP, which uses Monte Carlo tree search with particle filters to plan in large POMDPs without an explicit model. Sunberg and Kochenderfer [7] extend this to continuous observation spaces via double progressive widening. The MCTS

implementation incorporates observation progressive widening from POMCPOW, tree reuse across steps (ABT-style), and progressive bias initialization inspired by AlphaGo [8].

III. PROBLEM FORMULATION

A. POMDP Definition

The task is formalized as a POMDP $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, Z, R, \gamma)$.

State space $\mathcal{S}$. The full (hidden) system state is:

$$s = (\mathbf{q}, \mathbf{p}_{\text{blk}}, \mathbf{p}_{d_1}, \mathbf{p}_{d_2}, h), \quad (1)$$

where:

- $\mathbf{q} = [q_1, q_2, q_3, q_4, q_5, g] \in \mathbb{R}^6$ are the five joint angles (rad) and the binary gripper state $g \in \{0, 1\}$.
- $\mathbf{p}_{\text{blk}} = (x_b, y_b, \theta_b) \in \mathbb{R}^3$ is the pose of the red target block in the world frame.
- $\mathbf{p}_{d_1}, \mathbf{p}_{d_2} \in \mathbb{R}^3$ are the poses (x, y, θ) of the two blue distractor blocks.
- $h \in \{0, 1\}$ is a binary flag indicating whether the target block is currently held by the gripper.

The arm configuration $\mathbf{q}$ is fully observable (joint encoders). All block poses and h are *hidden*; the agent never directly observes the true positions of the target or distractors.

Action space $\mathcal{A}$. Actions are continuous six-dimensional vectors $a \in [-1, 1]^6$:

$$a = [\Delta q_1, \Delta q_2, \Delta q_3, \Delta q_4, \Delta q_5, g_{\text{cmd}}], \quad (2)$$

where Δq_i are joint velocity commands scaled to ± 0.05 rad/step, and $g_{\text{cmd}} \in [-1, 1]$ is the gripper command. Grasping is implemented as auto-grasp: when the end-effector enters a sphere of radius $r_{\text{grasp}} = 10$ mm around any block, that block is automatically attached to the gripper via a position constraint. The agent therefore controls approach and positioning rather than the grasp trigger directly.

Observation space $\mathcal{O}$. At each step the agent receives a partial observation $o_t \in \mathbb{R}^{15}$ (plain PPO) or $\mathbb{R}^{18}$ (belief PPO):

- $\mathbf{q} \in \mathbb{R}^6$: current joint angles (fully observed).
- $\hat{\mathbf{p}}_{\text{blk}_t} \in \mathbb{R}^3$: wrist camera estimate of the target position (x, y, θ) , corrupted by distance-dependent Gaussian noise (plain PPO) or replaced by the particle filter mean μ_t (belief PPO).
- $\sigma_t \in \mathbb{R}^3$: particle filter uncertainty $(\sigma_x, \sigma_y, \sigma_\theta)$ (belief PPO only).
- $\mathbf{p}_{\text{ee}} \in \mathbb{R}^3$: end-effector Cartesian position (x, y, z) from forward kinematics.
- $\mathbf{p}_{\text{goal}} \in \mathbb{R}^2$: fixed goal position (x, y) in the world frame.
- $h \in \{0, 1\}$: holding flag (1 if target is grasped).

Transition model T . Because the state space is continuous, the transition is not a matrix but a physics-defined mapping $s_{t+1} = f(s_t, a_t)$ implemented by the MuJoCo engine (mj_step, 10 substeps per action, $\Delta t = 0.002$ s). Joint angles update deterministically from velocity commands. Blocks are free rigid bodies under gravity; they remain stationary unless in contact with the gripper. When $d_t < r_{\text{grasp}}$, a position constraint rigidly attaches the nearest block to the end-effector

with a fixed vertical offset; releasing the gripper drops the constraint and the block falls under gravity.

Observation function Z . The wrist camera observation of the target is:

$$P(o_t | s_t) = \begin{cases} \mathcal{N}(\mathbf{p}_{\text{blk}}^{xy\theta}, \sigma_{\text{obs}}(d_t)^2 \mathbf{I}) & \text{target visible} \\ \delta(o_t - o_\tau) & \text{target occluded} \end{cases} \quad (3)$$

where $\sigma_{\text{obs}}(d_t) = \sigma_{\text{ep}} \cdot (d_t/d_{\text{ref}})$ is the distance-dependent noise standard deviation, σ_{ep} is fixed per training run, $d_{\text{ref}} = 0.15$ m, and o_τ is the last visible reading replayed during occlusion.

Reward R and discount γ . The reward function is detailed in Table I. The discount factor is $\gamma = 0.99$ with a maximum episode horizon of $T = 200$ steps, after which the episode terminates as a timeout failure.

B. Sources of Partial Observability

Three distinct uncertainty sources make this task a POMDP rather than an MDP. Each affects the observation function Z differently and requires a different handling strategy.

1. Distance-dependent sensor noise. The wrist camera returns a Gaussian-corrupted estimate of the target's (x, y, θ) position with standard deviation $\sigma_{\text{obs}}(d_t) = \sigma_{\text{ep}} \cdot (d_t/d_{\text{ref}})$, where d_t is the current end-effector-to-target distance and $d_{\text{ref}} = 0.15$ m. Crucially, this uncertainty is *reducible*: as the agent approaches the target, d_t decreases and the observation becomes more accurate. This creates an implicit information-gathering incentive: the agent must move closer to localize reliably before grasping, without any explicit active perception reward.

2. Occlusion-induced observation dropout. When a distractor block passes between the wrist camera and the target, the observation is entirely absent for that step. The agent receives no update to Z during occlusion; it must rely on its last known estimate or, in the belief PPO case, the particle filter's prediction step (which correctly grows uncertainty during the gap). Per-step occlusion rates reach 48% during early training exploration, meaning the agent frequently acts on stale information for extended periods. Unlike sensor noise, occlusion-induced uncertainty is *not reducible by approaching*; it is resolved only by repositioning to clear the distractor from the camera's line of sight.

3. Distractor identity ambiguity. Both the target and distractor block poses are hidden states in s ; the wrist camera observation covers only the target block and gives no direct information about distractor positions. Because the auto-grasp threshold ($r_{\text{grasp}} = 10$ mm) is comparable to the mean inter-block spacing (~ 17 mm), a position estimate corrupted by noise or occlusion can place the agent within grasp range of a distractor rather than the target. A wrong grasp terminates the episode with a -30 penalty. This uncertainty source is a direct consequence of the first two: accurate, unoccluded localization eliminates wrong-grasp risk; degraded localization makes it non-negligible.

Taken together, these three sources create a genuine exploration–exploitation tension: the agent must decide

TABLE I
REWARD FUNCTION COMPONENTS. ALL DISTANCE THRESHOLDS ARE IN MM.

Component	Value	Condition
Step cost	-1.0	Every step
XY approach	$+3.0 \cdot \Delta d_{xy}$	Per step, potential-based
Proximity bonus	$+2.0(1 - d/25\text{mm})$	$d_{xy} < 25$ mm
Reach block	$+5.0$ (once)	$d_{xy} < 15$ mm
Grasp success	$+10.0$	Auto-grasp, target
Grasp fail	-2.0	Auto-grasp misses
Distractor grasp	$-30.0 + \text{terminate}$	Wrong block
Carry shaping	$+5.0 \cdot \Delta d_{\text{goal}}$	While holding
Near goal	$+5.0$ (once)	Holding, $d_{\text{goal}} < 30$ mm
Perfect place	$+50.0$	$d_{\text{goal}} < 10$ mm
Precise place	$+30.0$	$d_{\text{goal}} < 20$ mm
Close place	$+10.0$	$d_{\text{goal}} < 40$ mm
Miss	-10.0	Release outside goal

whether to spend steps reducing uncertainty (approaching for a better sensor reading, or repositioning to clear an occlusion) or to act on its current, imperfect belief and risk a failed grasp.

C. Reward Function

The reward function uses potential-based shaping [5] for approach and carry, one-time milestone bonuses to avoid per-step exploitation, and large terminal rewards to ensure the overall objective dominates. The full breakdown is given in Table I.

D. Simulation Environment

Figure 1 depicts the MuJoCo pick-and-place workspace. The SO-ARM101 is a 5-DOF desktop arm with a parallel-jaw gripper. The MuJoCo scene contains a red 2×4 LEGO brick (the target, 32×16 mm footprint), two blue 2×2 LEGO bricks (distractors, 40×20 mm footprint, $\sim 25\%$ larger to exaggerate occlusion), and a fixed yellow goal marker approximately 15 cm from the home configuration. Distractors are spawned uniformly within 25 mm of the target at each episode reset, producing an inter-block spacing of 17 mm on average and ensuring that $\sim 34\%$ of distractors are within the grasp threshold of the target.

Per-step occlusion rates (measured across 86% of episodes) sit near 16% under a trained policy rollout, rising to 48% under random exploration. This ensures that belief tracking provides meaningful benefit without making the task unsolvable.

IV. SOLUTION APPROACH

A. Plain PPO (Baseline)

The plain PPO agent treats the POMDP as a standard MDP by using the stale sensor model: during occlusion, the policy receives the cached last-visible wrist observation o_τ as if it were the current reading. The 15-dimensional policy observation is:

$$\phi_{\text{plain}} = [\mathbf{q}(6), \hat{\mathbf{p}}_{\text{blk}(3)}^{xy\theta}, \mathbf{p}_{\text{ee}}(3), \mathbf{p}_{\text{goal}}(2), h_{(1)}], \quad (4)$$

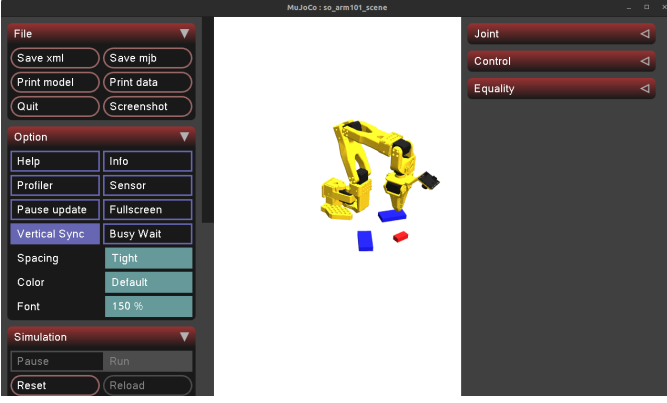


Fig. 1. MuJoCo scene (so_arm101_scene): SO-ARM101 with the red target brick, two blue distractors, and workspace used for training.

where $\mathbf{p}_{\text{blk}}^{\hat{}}$ is the stale/noisy wrist reading, $\mathbf{p}_{\text{ee}}$ is the end-effector Cartesian position (from forward kinematics), and $\mathbf{p}_{\text{goal}}$ is the fixed goal position in the world frame. The policy is an MLP with two hidden layers of 256 units each (ReLU activations, linear output), matching the SB3 PPO default architecture.

Training uses the SB3 PPO implementation [9] with $n_{\text{env}} = 3$ parallel environments, $n_{\text{steps}} = 2048$, entropy coefficient 0.05, $\gamma = 0.99$, clip ratio 0.2, and VecNormalize for both observations and rewards. The plain agent is trained for 5M environment steps; the best checkpoint is saved by an evaluation callback (50 evaluation episodes every 10k steps).

B. Belief-Augmented PPO

1) *Particle Filter Belief Tracker*: A bootstrap particle filter is maintained [3] over the target block’s hidden pose $\mathbf{p}_{\text{blk}} = (x, y, \theta)$. The filter state is $N = 300$ weighted particles $\{(x^i, w^i)\}_{i=1}^N$ updated at each step as follows.

Predict. Each particle is perturbed with Gaussian process noise to account for residual uncertainty in the block’s rest position:

$$x_{t|t-1}^i = x_{t-1}^i + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \text{diag}(\sigma_p^2, \sigma_p^2, \sigma_{p\theta}^2)), \quad (5)$$

with $\sigma_p = 1$ mm/step and $\sigma_{p\theta} = 0.01$ rad/step. The small process noise reflects that the blocks are stationary objects that do not move spontaneously.

Update. When a wrist observation o_t is available (target not occluded), particle weights are updated by the Gaussian likelihood:

$$w^i \propto w^i \cdot \mathcal{N}(o_t^{xy}; x_{xy}^i, \sigma_{\text{lik}}^2 \mathbf{I}), \quad (6)$$

where $\sigma_{\text{lik}} = 1.5 \sigma_{\text{obs}}(d_t)$. The widened likelihood ($1.5\times$) prevents particle impoverishment: using the raw sensor sigma collapses the effective sample size $N_{\text{eff}} = 1/\sum_i (w^i)^2$ to near 1 each step, degenerating the filter into a noisy running average. During occlusion, the update step is skipped entirely; particles spread freely via the predict step, correctly growing uncertainty without incorporating spurious information.

Resample. Systematic resampling is applied when $N_{\text{eff}} < N/2$. After resampling, 5% of particles are re-injected near the last visible observation to prevent filter degeneracy during extended occlusion.

Belief summary. The filter’s output is compressed to $\hat{\mathbf{b}}_t = (\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t) \in \mathbb{R}^6$:

$$\boldsymbol{\mu}_t = \sum_i w^i x^i, \quad \boldsymbol{\sigma}_{t,k} = \sqrt{\sum_i w^i (x_k^i - \mu_{t,k})^2}, \quad (7)$$

for each dimension $k \in \{x, y, \theta\}$. The standard deviation $\boldsymbol{\sigma}$ summarizes how uncertain the filter is about the target’s position: high $\boldsymbol{\sigma}$ indicates recent or extended occlusion, while low $\boldsymbol{\sigma}$ indicates confident localization.

2) *Policy Observation and Architecture*: The belief-augmented policy receives an 18-dimensional observation:

$$\phi_{\text{belief}} = [\mathbf{q}_{(6)}, \boldsymbol{\mu}_{(3)}, \boldsymbol{\sigma}_{(3)}, \mathbf{p}_{\text{ee}(3)}, \mathbf{p}_{\text{goal}(2)}, h_{(1)}]. \quad (8)$$

The critical difference from plain PPO is the inclusion of $\boldsymbol{\sigma}$: the policy can learn to slow its approach when uncertainty is high (reducing the risk of triggering auto-grasp on a distractor at the wrong position) and act more aggressively when the filter is confident. No explicit information-gathering reward is used; this behavior must emerge from the task reward alone.

The reward for belief PPO uses the particle filter mean $\boldsymbol{\mu}$ in place of the raw sensor reading when computing distance-based shaping terms. This ensures that the reward gradient is also filtered, providing a cleaner learning signal at high noise levels.

All hyperparameters are identical to plain PPO, except the belief agent trains for 6.4M steps to account for the larger and harder-to-exploit observation space. The policy network architecture is the same MLP [256, 256] as the baseline.

V. RESULTS

A. Training Convergence

Fig. 4 shows training curves for both agents. Both discover the approach-shaping gradient around 200k steps, after which reward rises monotonically until plateauing. Plain PPO reaches its best checkpoint at step 3.74M; belief PPO plateaus later, reaching its best checkpoint at 6.37M steps. The slower convergence of belief PPO is expected: the 18-dimensional observation contains the $\boldsymbol{\sigma}$ uncertainty signal, whose relevance to the task reward requires more experience to exploit. Neither agent exhibits catastrophic forgetting after the best checkpoint, indicating stable training.

B. Noise Sweep: Main Result

Both best-model checkpoints are evaluated across eight noise levels ($\sigma_{\text{ep}} \in \{3, 5, 8, 10, 13, 16, 20, 30\}$ mm) with 50 evaluation episodes each at fixed seed 42. Results are shown in Fig. 5 and Table II.

The results reveal a clear noise-regime structure. At $\sigma_{\text{ep}} = 3$ mm, plain PPO achieves 66% success with mean episode length 111 steps, while belief PPO manages only 36% at 196 steps. The plain agent acts decisively because its wrist

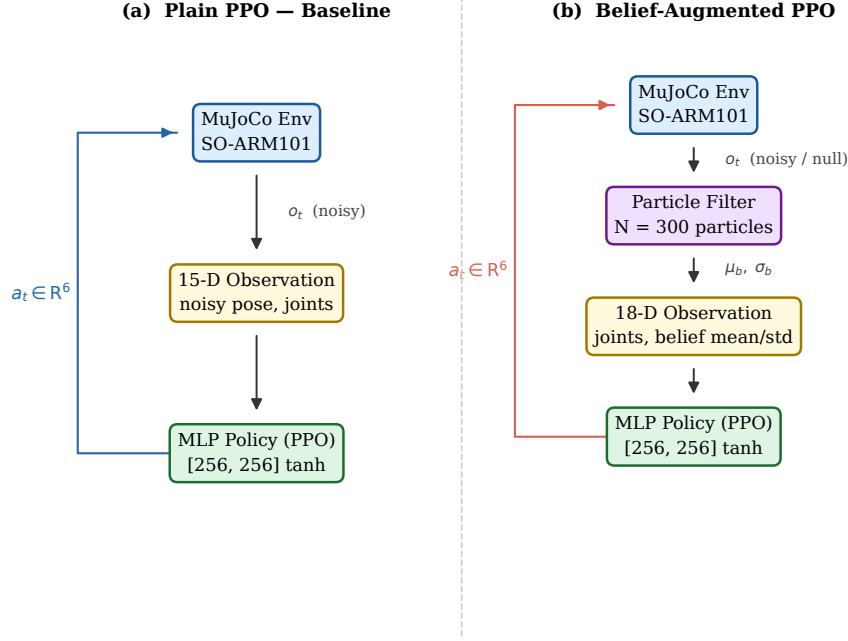


Fig. 2. Architecture comparison. (a) Plain PPO exposes noisy (or stale, during occlusion) sensor readings as a 15-D vector directly to the policy. (b) Belief-augmented PPO routes each observation through a 300-particle bootstrap filter; the filter summary (μ_b, σ_b) augments the 18-D policy input so the agent can reason about its own localization uncertainty.

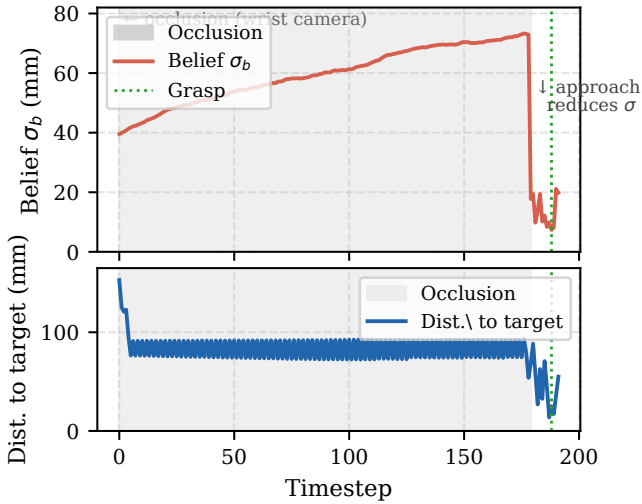


Fig. 3. Belief uncertainty σ_b (top) and distance to target (bottom) over one representative successful episode. The shaded region marks prolonged wrist-camera occlusion (steps 0–178): the particle filter’s uncertainty grows steadily as no measurements arrive. Once occlusion clears, the robot approaches and the distance-dependent noise model $\sigma_{\text{obs}} \propto d_t$ causes σ_b to collapse from 73 mm to < 10 mm within a few steps. The dashed line marks the successful auto-grasp at step 188.

observations are already accurate (~ 3 mm error); the belief agent is overly cautious because its particle filter is initialized with spread σ_{ep} at episode start, leading the policy to perceive uncertainty even when none is warranted.

In the moderate regime ($\sigma_{\text{ep}} = 5\text{--}20$ mm), belief PPO

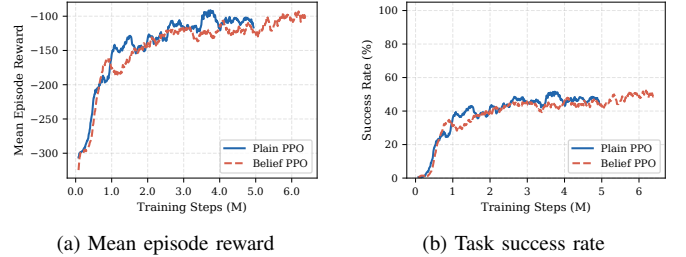


Fig. 4. Training curves (15-step moving average). Success rate: fraction of evaluation episodes completing in < 200 steps. Belief PPO converges ~ 1.6 M steps later but reaches a comparable peak.

TABLE II
NOISE SWEEP RESULTS (50 EPISODES PER LEVEL, SEED 42). **BOLD** = BETTER. Δ = BELIEF – PLAIN.

σ_{obs} (mm)	Plain (%) $\uparrow$	Belief (%) $\uparrow$	Δ (%)
3	66	36	−30
5	46	52	+6
8	58	60	+2
10	34	58	+24
13	48	48	0
16	44	48	+4
20	40	46	+6
30	42	32	−10

wins or ties at every level, with the largest advantage at 10 mm (+24%, belief 58% vs. plain 34%). At this noise level, the particle filter converges within a few visible observations (typically < 5 steps) to a mean error of ~ 4 mm, well below

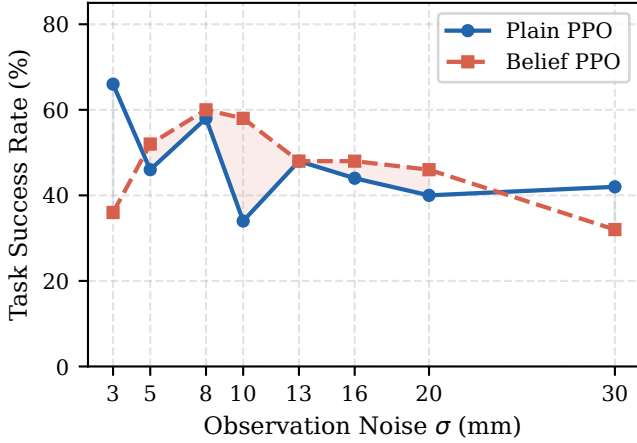


Fig. 5. Task success rate vs. σ_{obs} . Shaded regions show which method leads. Belief PPO outperforms plain PPO across the moderate-noise regime (5–20 mm); plain PPO dominates at very low noise (3 mm).

the grasp threshold, while the plain agent’s stale observation carries 10 mm error, enough to cause localization failures during occlusion.

At $\sigma_{\text{ep}} = 30$ mm (beyond the training distribution), both methods degrade: plain PPO to 42% (stale readings too inaccurate), belief PPO to 32% (particle impoverishment: the likelihood function is so flat at $\sigma_{\text{lik}} = 45$ mm that particle weights barely update, preventing convergence).

The $\sigma_{\text{ep}} \approx 8$ mm crossover is theoretically consistent with the signal-to-noise structure of the bootstrap filter: below it, the raw observation is already close to optimal and the $O(N)$ filter overhead is unjustified; above it, filtering reduces position error enough to consistently reach the 10 mm grasp sphere before timeout.

C. Failure Mode Analysis

Across all noise levels, task success rate and grasp rate are nearly identical ($\text{success\%} \approx \text{grasped\%}$), with zero wrong-block grasps recorded for either agent. This reveals that the bottleneck is *localization and approach*, not manipulation skill: once the agent grasps the target block, it reliably carries and places it (placement phase never fails). All failures ($\sim 60\%$ of episodes at mid-to-high noise) are pre-grasp timeouts in which the agent could not navigate to within 10 mm of the target within the 200-step horizon.

The absence of distractor confusion (zero wrong grasps) is notable given that 34% of distractor spawns place a distractor within the grasp threshold of the target. Both agents learned to approach the target’s estimated position accurately enough that auto-grasp always activates on the correct block. This is consistent with the high placement success rate conditional on grasping: both agents learned precise approach behavior; their failure mode is navigation under uncertainty, not discrimination.

TABLE III
POMCP-MCTS VS. TRAINED POLICIES. PPO COLUMNS SHOW RESULTS AT THE TRAINING NOISE LEVEL ($\sigma_{\text{ep}} = 5$ mm). POMCP RUNS AT FIXED $\sigma_{\text{ep}} = 5$ mm (TRAINING NOISE OF BELIEF PPO).

Method	Success (%) $\uparrow$	Mean Steps $\downarrow$	Plan time
Plain PPO	46	~ 168	instant
Belief PPO	52	~ 150	instant
POMCP-MCTS	60	143	28.1 s/step

D. POMCP Stretch Goal

As a stretch goal, POMCP-MCTS is evaluated with the direct MuJoCo simulator as the generative model. The planner uses UCB1 tree search with 200 simulations per step, a heuristic rollout policy, observation progressive widening ($k = 2$, $\alpha = 0.5$), ABT-style tree reuse across steps, and progressive bias initialization (heuristic action seeded at $Q = 0$; all others at $Q = -40$). The carry-over tree is reset every 40 steps to prevent Q-value corruption on stuck episodes. Results on the same 20-episode evaluation (Table III) show 60% success at a mean planning cost of 28.1 s/step and 143 mean steps per episode.

POMCP-MCTS shows a bimodal outcome distribution: successful episodes complete in 16–86 steps (median ~ 40), while the 8 failures all hit the 300-step timeout. This brittleness reflects a known limitation of tree search under a small simulation budget: with 200 simulations per step, Q-value estimates carry a standard error of ~ 12 reward units, insufficient to reliably distinguish between actions whose true values differ by 5–15 reward. The 300-step timeout failures correspond to hard initial configurations where the heuristic rollout policy is itself insufficient, and POMCP inherits this weakness through progressive bias initialization. Despite the higher nominal success rate, the 28.1 s/step planning cost makes POMCP impractical for real-time deployment; trained policies achieve comparable or better success at negligible inference cost.

VI. DISCUSSION AND CONCLUSION

These results show that particle filter belief augmentation provides a meaningful performance advantage when observation noise is in a regime where filtering is effective, specifically when σ_{ep} is large enough that the raw sensor reading carries significant error, but not so large that the filter itself cannot converge. The crossover at ~ 8 mm and the degradation at 30 mm are both theoretically motivated consequences of the bootstrap filter’s convergence characteristics. The practical implication is that belief augmentation is not universally beneficial: in low-noise settings, the cost of filter initialization and the induced caution in the policy can outweigh the smoothing benefit. Practitioners should choose between a stale-sensor baseline and a belief-augmented agent based on the expected operating noise level.

On reward engineering, developing a stable reward function for multi-phase manipulation required iterating through

18 training runs and discovering five distinct classes of reward hacking. Per-step state-occupancy rewards (“be near the block”, “keep gripper closed near block”, “be near goal while holding”) were universally exploited: the agent learned to enter the rewarded state and remain there indefinitely, collecting reward without task progress. The fix in every case was to convert the reward to either a one-time milestone (fires once per episode on the first entry into the state) or a potential-based shaping term (rewards only the improvement in a scalar function, not the absolute value). These are precisely the conditions under which Ng et al. [5] guarantee that the optimal policy is preserved. The grasp-attempt spamming exploit in particular (where the agent discovered it could repeatedly attempt grasps from an incorrect height and collect positive rewards for each attempt) required four separate design iterations to resolve robustly.

Regarding limitations and future work, the current environment uses a 2D grasp check (XY proximity only) rather than a full 3D check, a pragmatic choice after six failed training runs with the physically correct 3D threshold. The issue was the lack of a gradient for the Z descent phase: with a 3D grasp condition, the agent learns to approach in XY but finds no incentive to descend to table height, since the 3D proximity reward is zero when hovering directly above the block. Future work could address this with a curriculum that first rewards XY approach, then adds a Z descent milestone only after XY convergence is stable. Additionally, real robot evaluation via the ROS2 integration (`policy_runner.py`) was begun but not completed due to time constraints; sim-to-real transfer remains an open question for this platform. The POMCP-MCTS results highlight that a larger simulation budget (~ 2000 simulations/step) would be needed for reliable Q-value convergence; future work could explore learned rollout policies to improve planning efficiency.

This work demonstrated that belief-augmented reinforcement learning with a bootstrap particle filter improves pick-and-place success under observation noise and occlusion in the moderate-noise regime ($\sigma_{\text{ep}} = 5\text{--}20$ mm), with a peak advantage of +24% at $\sigma_{\text{ep}} = 10$ mm. The advantage reverses at very low noise, where raw observations are already accurate, and at very high noise, where the particle filter degrades. These findings provide a practical characterization of when belief augmentation is worth its added complexity, and the failure-mode analysis identifies localization (not manipulation) as the primary performance bottleneck in the POMDP pick-and-place setting.

CONTRIBUTIONS AND RELEASE

The project was designed and built the MuJoCo simulation environment from scratch (`lego_pick_env.py`), including the reward function, occlusion model, constraint-based grasping, and the VecNormalize training pipeline. The bootstrap particle filter (`particle_filter.py`) and the belief-augmented observation wrapper were implemented independently. The POMCP direct planner and

MCTS variant (including tree reuse, observation progressive widening, periodic reset, and progressive bias initialization) were implemented from scratch (`train_pomcp.py`, `train_pomcp_mcts.py`). The PPO learning algorithm itself is from Stable-Baselines3 [9]; the MuJoCo physics engine and SO-ARM101 robot assets are third-party.

The author grant permission for this report to be posted publicly.

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